

Cadi Ayyad University

Higher School of Technology of Safi

Computer Science Department • Web Techniques

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Session 5 — Introduction to PHP and Dynamic Web Pages

Objective: Students discover PHP server-side scripting: variables, echo, conditions, and loops.

1. Key Concepts

What is PHP?

PHP (Hypertext Preprocessor) is a **server-side** scripting language. Unlike HTML/CSS which run in the browser, PHP code executes on the web server. The server runs the PHP, generates HTML, and sends *only* the HTML to the browser. The user never sees the PHP source code.

PHP Tags and Syntax

```
<?php
    // This is a comment
    $name = 'Safi';           // string variable
    $year  = 2026;           // integer variable
    $gpa   = 15.75;          // float variable
    echo 'City: ' . $name;    // concatenate with dot (.)
?>
```

- All variables start with \$.
- echo outputs content to the browser.
- Statements end with a semicolon ;.
- Strings use single or double quotes. Concatenation uses the dot operator ..

Conditions

```
<?php
    $score = 14;
    if ($score >= 10) {
        echo 'Passed (' . $score . '/20)';
    } elseif ($score >= 8) {
        echo 'Borderline';
    } else {
        echo 'Failed';
    }
?>
```

Loops

```

<?php
    // for loop: count from 1 to 5
    for ($i = 1; $i <= 5; $i++) {
        echo $i . ' ';
    }
    // Output: 1 2 3 4 5

    // while loop: repeat while condition is true
    $x = 10;
    while ($x > 0) {
        echo $x . ' ';
        $x = $x - 2;
    }
    // Output: 10 8 6 4 2
?>

```

2. Guided Examples

Example A — Mixing PHP and HTML

```

<?php $module = 'Web Techniques'; $students = 32; ?>
<!DOCTYPE html>
<html><body>
    <h1><?php echo $module; ?></h1>
    <p>Enrolled students: <?php echo $students; ?></p>
    <?php
        if ($students > 30) {
            echo '<p><b>Large class!</b></p>';
        }
    ?>
</body></html>

```

Example B — for loop generating HTML

```

<?php
    echo '<ul>';
    for ($i = 1; $i <= 4; $i++) {
        echo '<li>Student ' . $i . '</li>';
    }
    echo '</ul>';
?>

```

Output: an unordered list with items Student 1, Student 2, Student 3, Student 4.

3. Exercises

Exercise 1 — Trace the Loop — Step by Step

For the PHP loop below, complete the trace table by filling in the value of `$i` and `$total` after each iteration. Then write the final output.

```
<?php
    $total = 0;
    for ($i = 1; $i <= 5; $i++) {
        $total = $total + $i;
    }
    echo 'Sum = ' . $total;
?>
```

Iteration	\$i	\$total (after step)	Condition (\$i <= 5)
1			
2			
3			
4			
5			
End			False — loop exits

Final output: _____

Exercise 2 — Predict the Output

Write what each PHP snippet will output in the browser. Show your reasoning.

Snippet A:

```
<?php
    $a = 4; $b = 7;
    echo ($a * $b) . ' is ' . ($a * $b > 25 ? 'big' : 'small');
?>
```

Output: _____

Snippet B:

```
<?php
    $n = 3;
    $result = '';
    while ($n <= 12) {
        $result .= $n . '-';
        $n += 3;
    }
    echo $result;
?>
```

Output: _____

Exercise 3 — Write a PHP Function

In PHP, you can define reusable functions just as in JavaScript. Below is the syntax; write the function described.

```
function functionName($param1, $param2) {
    // body
    return result;
}
```

Write a PHP function called `letterGrade` that:

- Takes one parameter: `$score` (a number out of 20).
- Returns 'A' if `score >= 16`.
- Returns 'B' if `score >= 12`.
- Returns 'C' if `score >= 10`.
- Returns 'F' otherwise.

Then write the echo statement that calls it with `score = 13` and shows the result.

Exercise 4 — Mixed HTML and PHP — Complete the Page

Complete the missing PHP code in the blanks so that the page works as described in the comments.

```
<?php
    $student = 'Khadija';
    $score    = 11;
    $max      = 20;
?>
<!DOCTYPE html><html><body>
    <h1>Result for <?php echo _____ ; ?></h1>
    <p>Score: <?php echo _____ . '/' . _____ ; ?></p>
    <?php
        if ( _____ >= 10) {
            echo '<p style="color:green">PASSED</p>';
        } else {
            echo _____ ;    // print FAILED in red
        }
    ?>
</body></html>
```

Exercise 5 — PHP Debugging — Find 5 Errors

The following PHP script contains **5 errors** (syntax, logic, and type errors). Identify each error, state the line, explain what is wrong, and write the fix.

```
<?php
    name = 'Ali';
    $age = '25';
    echo 'Hello ' . name . ', you are ' . $age . ' years old.';

    if $age > 18 {
        echo 'Adult';
    } else
        echo 'Minor'
    }

    for ($i == 0; $i < 3; $i++) {
        echo $i;
    }
?>
```

Error 1 (line __): _____

Error 2 (line __): _____

Error 3 (line __): _____

Error 4 (line ____): _____

Error 5 (line ____): _____

4. Session Recap

PHP is a server-side language: the server runs it and sends HTML to the browser. Variables begin with \$ and hold strings, numbers, or booleans. echo outputs content; the dot (.) concatenates strings. if/elseif/else handles decision branches. for and while loops repeat code a controlled number of times. PHP can generate HTML dynamically — making pages interactive and data-driven. Next session: JavaScript, which runs on the client side.